

Preparation of beeswax/multi-walled carbon nanotubes as novel shape-stable nanocomposite phase-change material for thermal energy storage

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ABSTRACT

Development of phase-change material (PCM) as thermal energy storage for building envelopes is promising for energy utilization. However, there are two major drawbacks of PCM application, which are low thermal conductivity and high-volume reduction due to phase-change transition. One solution is to develop a shape-stabilized phase-change material (SSPCM) as a composite that is able to prevent leakage during the transition from solid to liquid. Therefore, the objective of this study is to prepare beeswax/multi-walled carbon nanotubes as form-stable nanocomposite phase-change material for thermal energy storage, based on previously unattempted methods. Beeswax was being used as PCM because of its high latent heat and multi-walled carbon nanotubes (MWCNTs) as supporting material with high thermal conductivity. There are three types of MWCNTs applied in this research: pristine MWCNTs, ball-milled MWCNTs and acid-treated MWCNTs. Beeswax/CNT composite samples were prepared with ratios of 5 and 20 wt%. Composite samples were tested from structure modification and thermal performance, including latent heat, sensible heat, melting point, solidifying point, thermal conductivity, and thermal-cycle testing for up to 300 cycles. Experimental results showed that thermal conductivity of novel shape-stable nanocomposite PCM increased by a factor of 2 and there was no significant phase transition in the melting or solidifying temperature. The high heat storage capability and thermal conductivity of nanocomposite PCM enable it to be a potential material for thermal energy storage in practical applications.

1. Introduction

Sustainability engineers are focused on addressing increasing energy demands while reducing greenhouse gas emissions from the use of fossil fuels. Energy storage is a crucial part of any next-generation energy distribution system. Energy consumption in buildings accounts for about 30% of global energy and greenhouse gas emissions [1]. This share is expected to increase significantly, as people spend more time indoors. Heating, Ventilation, and Air Conditioning (HVAC) accounts for the greatest share of the building's total energy consumption, and contributes to about 50% of building emissions in developed nations [2,3]. Thermal energy storage (TES) is very effective methods for improving energy savings, energy efficiency, and ensuring full utilization of renewable energy [4–7]. Thermal energy can be kept as sensible heat, latent heat, thermo-chemical heat, or a combination of these. From the all TES methods, latent heat thermal energy storage (LHTES)

using phase-change material (PCM) is preferred because of its high-storage density and characteristic of storing heat at a constant temperature. Many PCMs are available for utilization in LHTES applications, such as paraffin, non-paraffin organic acids, salt hydrates, clathrates, and eutectic organic and inorganic compounds [8–13].

Latent-heat energy storage has been applied widely. Cabeza et al. [14] listed PCM properties in which the application settings are divided by its phase-change temperature. In order to apply in buildings, three temperature ranges were recommended: (i) up to 21 °C for cooling applications, (ii) 22–28 °C for human-comfort applications, and (iii) 29–60 °C for hot-water applications. The thermal energy storage method provides much higher energy-storage density with a smaller temperature fluctuate when compared with the sensible-heat-storage method. However, practical difficulties often arise in applications of LHTES due to the low thermal conductivity, density change, and stability of properties that can occur with extended cycling, and infrequent

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Fig. 1. Beeswax used in the experiment.

phase segregation and subcooling of the phase-change materials.

A great deal of candidate inorganic and organic PCMs and their mixtures have been recently studied as LHTES materials that can be impregnated into common building materials [10,15–18]. Su-Gwang Jeong et al. investigated the composite formation of *silica fume* and inorganic PCM inorganic for concrete building materials [19]. Kong et al. emphasize energy storage for use with building solar panels using paraffin and expanded perlite [20]. Among the two types of PCMs, organic PCMs have been attracting great attention due to their good properties for example a suitable phase-change temperature, high latent-heat capacity, congruent melting, non-toxicity, no supercooling, low vapor pressure, little to no volume change during solid–liquid phase change, high thermal and chemical stability after a long-term utility period [21,22]. On the other hand, organic PCMs have low thermal conductivity at approximately 0.1–0.3 W/m K [9,10] which limits their applications.

A popular organic PCM is beeswax, which is wide-spread globally and is used for a range of applications [23,24]. Beeswax is comprised of esters of fatty acids and long-chain alcohols. It falls under the category of organic non-paraffin PCMs. The empirical formula of beeswax is $C_{15}H_{31}COOC_{30}H_{61}$. Beeswax commonly consumed in beauty products, food grade finishes, metal handling, etc. Its development as PCM has been reviewed by Sinaringati, et al. for use as the heat source in an

infant incubator [25]. Amin, et al. 2017 tested the modification of nano-PCM using beeswax and graphene with various mass ratios to determine the thermal characteristics. The thermal conductivity of this nano-PCM increased by 12% [26].

Nevertheless, beeswax possesses some drawbacks that hinder its use as an organic PCM, including low thermal conductivity and liquid leakage during the phase transition. Several strategies for these difficulties have been studied, such as encapsulation and shape stabilization [27]. Currently, form-stable PCM composites with good thermal conductors for supporting materials can provide a solid shape even if the temperature is more than the melting point of the PCM. These materials are referred to as shape-stabilized phase-change materials (SSPCMs). Several studies have focused on the preparation and characterization of SSPCMs using porous and carbon-based materials, including graphite, carbon nanofibers, carbon nanotubes, graphite nanoplatelets, graphene oxide, and others. This modification of the material effectively resolves the problems of leakage and low thermal conductivity because carbon has a high thermal conductivity of 2000–6000 W/m K [21].

However, composite PCMs prepared using this method need to be form-stable over many melting-freezing cycles in order to prevent leakage when the PCM is in its melted state. Qian et al. [28] reported that after 200 cycles, a composite of polyethylene glycol (PEG) and porous diatomite remained stable. Mehrali et al. [29] studied the addition of carbon nanospheres (CNS) in stearic acid (SA) to form a composite, with a weight ratio of PCM 80%. Thermal characterization measurements after 1000 cycles indicated no significant change. Guo, et al. [30] compared carbon added to paraffin for structural support and found that it significantly increased the thermal conductivity and molecular interaction of the composite. Min et al. [31] studied the formation of PEG as an SSPCM with porous silica sphere radicals for TES applications. Tang et al. [32] modified a novel eutectic PCM MA-SA with CNT in a composite and found that the highest thermal conductivity was obtained when the PCM was added at 15 wt% CNT. Several experiments have been conducted at a more general level to evaluate composites' potential for enhancing the energy efficiency of buildings. Utilization of composites during winter was analyzed by Tyagia, et al. [33] with the composite attached into building envelopes of concrete walls.

This study focused on the preparation of beeswax and Multi-Walled Carbon Nanotubes (MWCNT) for use as a novel form-stable nano-composite PCM. Beeswax with a melting point of 62.3 °C [26] has not been used in an SSPCM yet. Characterizations of the PCM composite were conducted using SEM, FT-IR and DSC analysis for the

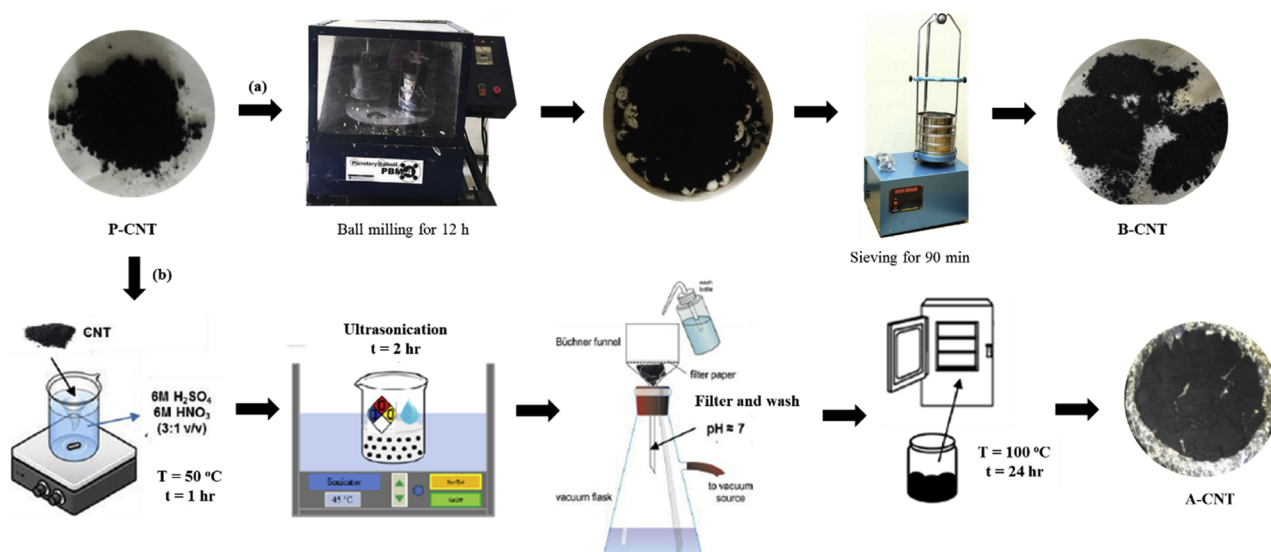


Fig. 2. CNT treatment methods: (a) Ball milling (b) Acid oxidation.

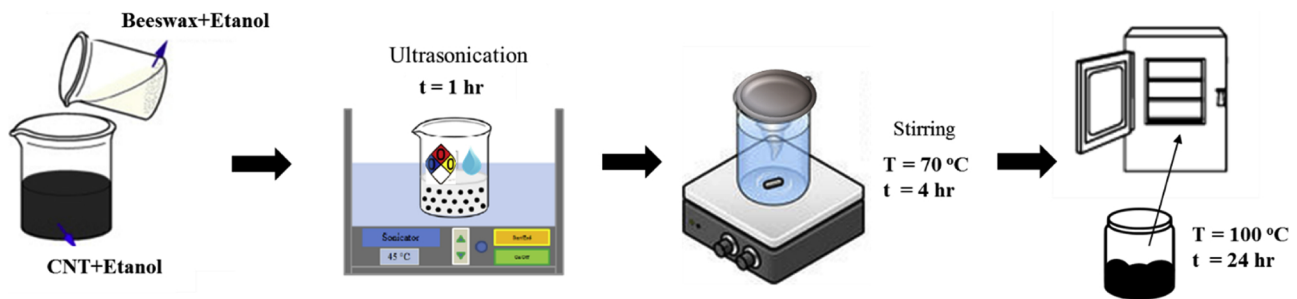


Fig. 3. Composites preparation by impregnation.

Table 1

Phase changing process for a period of time.

Sample	1 minute	5 minutes	10 minutes	15 minutes
Pure Beeswax				
Beeswax / P-CNT 5 wt. %				
Beeswax / B-CNT 5 wt. %				
Beeswax / B-CNT 20 wt. %				
Beeswax / A-CNT 5 wt. %				
Beeswax / A-CNT 20 wt. %				

determination of thermal properties and thermal reliability. The thermal conductivity of the SSPCM was enhanced by adding MWCNT which is high thermal conductivity. Furthermore, the stability of the SSPCM was also investigated by experimentally using thermal-cycle testing for up to 300 cycles. As literatures review, this research is the first reported about the application of beeswax and MWCNTs as an SSPCM for thermal energy storage.

2. Material and methods

2.1. Materials

The samples tested in this research was prepared from beeswax and MWCNTs. Several materials were purchased from Merck Millipore Co. Ltd.: beeswax with melting temperature of 60–62 °C, concentrated

nitric acid (HNO₃, 65 wt.%), concentrated sulfuric acid (H₂SO₄, 98 wt. %), and absolute ethanol. The density of the purchased beeswax is 820 kg/m³. All chemical reagents were analytical-grade and were used without further purification. Pristine MWCNTs (P-CNTs) with diameter in 10–30 nm and length 10–30 μm (special surface area larger than 60 m²/g, purity: 95 wt.%) were purchased from Chengdu Organic Chemical Co., Ltd. The material used is pictured in Fig. 1.

2.2. Methods

2.2.1. Preparation of MWCNTs

Two different methods were conducted to prepare the P-CNTs, namely ball milling and acid oxidation. Fig. 2 shows schematics diagram of the two methods. The P-CNTs were directly ball-milled for 12 h at a frequency of 14 Hz in a planetary milling machine without any

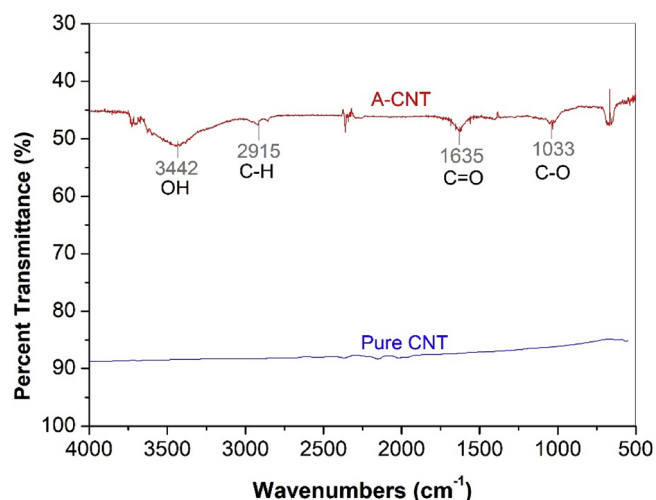


Fig. 4. FT-IR spectra of untreated and treated CNTs.

additives [34]. The ratio of ceramic balls and added MWCNTs was 50:1. Then, the treated MWCNTs were sieved with a 400-mesh screen for 90 min to remove macroscale contaminants. The MWCNTs prepared with ball milling are referred to as B-CNTs.

In the acid-oxidation procedure, two grams of P-CNTs and 400 ml acid mixture are stirred and heated at 50 °C for 1 h. The acid mixture in this study consisted of concentrated nitric (6 M) and sulphuric acids (6 M) in a ratio of 1:3. The sample after the reaction was diluted with aquadest several times until the solution showed no acidity (pH = 6.5), then was filtrated using a polycarbonate membrane filter with pore diameter of 45 nm. The neutral MWCNTs were then collected and dried at 100 °C for 24 h to remove all water. The MWCNTs treated with this acid oxidation method are referred to as A-CNTs.

2.2.2. Preparation of beeswax/MWCNT composites

The beeswax/MWCNT composites were made by the impregnation method [35], as illustrated in Fig. 3. A fixed amount of MWCNTs/ethanol (1 g/150 mL) suspensions was mixed with a beeswax/ethanol solution containing different beeswax amounts to achieve beeswax/MWCNT suspensions with weight ratios of 5 and 20 (%wt/wt). Following ultrasonic handling for 30 min, the mixtures were sealed in a beaker using plastic wrap and stirred for 4 h at 70 °C in a water bath

[36]. Sonication was important in order to produce well-dispersed and homogeneous beeswax/MWCNT composites. Finally, ethanol was evaporated by vacuum drying at 80 °C for 24 h. In order to test the different properties of the beeswax/MWCNT composite, six samples of composites with different preparations of CNTs were prepared.

2.2.3. Characterization

Various analytical methods were applied to characterize the treated MWCNTs and composite samples. The microstructure of beeswax/MWCNT composites was observed with scanning electron microscopy (SEM: ZEISS, Germany) at room temperature. A SEM with an accelerating voltage of 12 kV and resolution of 2 nm was used. The samples were coated with a gold coating of a few nanometers in thickness for SEM measurements.

Fourier-transform infrared spectroscopy (FTIR: Shimadzu, Japan) was also used to monitor the change of chemical groups upon curing. Potassium bromide (KBr) discs were molded from powder to be used as background material. The beeswax/MWCNT composites were analyzed over the range of 550–4000 cm^{-1} .

The thermal properties of the beeswax/MWCNT composites, such as its melting and freezing temperature and latent-heat capacity, were measured using differential scanning calorimetry (DSC: 8000 Perkin Elmer). The latent-heat capacity of the composite samples was determined by numerical integration of the area under the peaks that represent the solid–solid and solid–liquid phase transitions [37]. Differential scanning calorimetry measurements were conducted at 5 °C/min heating and cooling rate and temperature ranges of 0–100 °C and 100 to 0 °C.

The shape-stabilized properties of the beeswax/MWCNT composites were characterized using a hot-stage and a digital camera [35]. All samples were positioned on a hot stage and heated to 100 °C, which is higher than the samples' melting temperature. The shape changes of the samples were observed via tracking photography.

Thermal conductivity was measured by the absolute technique steady-state method: comparative cut bar [38]. Using a commercially available thermal conductivity apparatus (Ogawa Seiki, Japan), the thermal conductivity of samples was calculated by measuring the temperature difference at some distance under the steady-state heat flow through the sample. Composite samples were shaped for this test using a steel cylinder mold with outer and inner thickness of 4 and 2 mm. These cylindrical samples were then inserted into the copper cylinder of the instrument.

Finally, to determine the thermal stability of beeswax/MWCNT

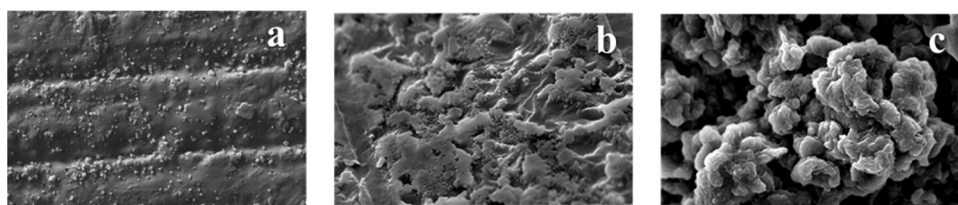


Fig. 5. SEM images (1000x) of: (a) Pure Beeswax (b) Beeswax/A-CNT 5 wt.% (c) Beeswax/A-CNT 20 wt.%.

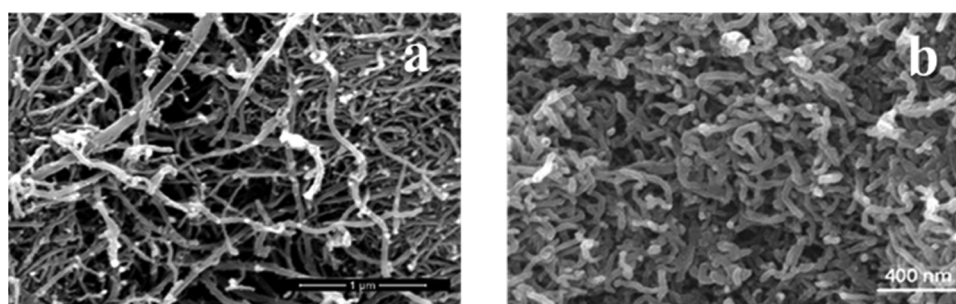


Fig. 6. SEM images of: (a) Untreated CNT [26] (b) Acid-treated CNT [27].

composite PCMs over many melting and freezing cycles, a thermal-cycling test was conducted using a system of accelerated thermal cycles. A hot plate was used to heat the sample in the laboratory, then the sample cooled and solidified, for up to 300 cycles. A thermocouple was embedded at the center of the samples for these experiments. The temperature and time transitions of the samples during these periods were recorded automatically using a data logger.

3. Results and discussion

3.1. Physical, chemical, and morphological properties

The observations of each sample were heated at 80 °C for five different amounts of time variations, namely 0 min. and 15 min. are listed in Table 1. The A-CNT composite maintains its shape without any lamination better than pure beeswax does. Composite with 20 wt% P-CNT is difficult to work with because it remains hard even after being heated up to its ignition point. The composite with 5 wt% A-CNT maintains its shape until the tenth minute and performs better than the composite with B-CNT.

Although the 5 wt% and 20 wt% B-CNT samples maintain their phase stability until the fifth minute of heating, both composites lose their form during the fifteenth minute. It seems that mechanical force does not change the CNT's chemical structure, but instead, the ball-milling process only increases the CNT's surface area [34].

However, the B-CNT composites are still more stable than pure beeswax. The greater the B-CNT content, the more stable the composite sample. Carbon nanotubes oxidized with a strong acid result are the best for use as an SSPCM. The acid-washed nanotubes trap beeswax inside the cavities, where it remains during the heating process.

Based on the physical examination, A-CNT composite tends to maintain its form due to changes in its chemical structure. Acid oxidation allows the CNTs to form into functional groups. The success of CNT oxidation can be observed from the FTIR spectra. The FTIR spectra shows absorption peaks that can be used to identify the functional groups in the samples. The desirable functional groups (–OH, C–O(H), and C=O) were observed. Comparison FTIR spectra of CNT and A-CNT precursors depicted in Fig. 4. The FTIR spectrum of pure CNT does not show any significant peak on the characteristic band, while several peaks appear in the A-CNT results (see Fig. 4).

The absorption peak at 1535 cm^{-1} was assigned for the carboxylic acid group (C=O) which is a specific carbonyl of the CNT. If there an absorption peak is more than 3200 cm^{-1} , there is a high chance that the CNT contains the hydroxyl group, thus the absorption peak at 3442 cm^{-1} indicates the existence of O–H bonds and hydrogen bonding. Peaks between $1750\text{--}1550\text{ cm}^{-1}$ indicate the C=O functional group which has alkanolic acid and C=C bonds. Peaks appearing between $3300\text{--}2700\text{ cm}^{-1}$ indicate C–H bonds at 2915 cm^{-1} . Lastly, the $1300\text{--}950\text{ cm}^{-1}$ wave indicates the C–O bond [39]. These functional groups strengthen the bond between beeswax and the acid-washed CNT, so that the sample will maintain its shape better.

The SEM images were taken to observe the morphology of the material structure before and after the tests. This observation was performed for pure beeswax and beeswax composite/A-CNT with mass ratios of 5 and 20 wt% because both of them successfully fabricated SSPCM. Therefore, the structure's form changes were observed further via SEM. It is shown in Fig. 5 (a) that the beeswax surface is flat and there were no lumps on its surface. Meanwhile, a pure CNT surface, as seen in Fig. 6 (a), looks like layers of long tubes, similar to the structure of fibers.

A strong acid-oxidized CNT yields A-CNT, which is a prerequisite component for forming SSPCM. A-CNT surface can be seen in Fig. 6 (b): it has shorter CNT tubes and it has a closer gap between tubes in comparison with that of pure CNT. This is mainly because of the decrease in CNT impurities as a result of strong acid oxidation. Less impurity indicates that the A-CNT is ready to be synthesized as a

composite by physical absorption into its pores [40].

Increasing the percentage of added A-CNT affects the composite's microstructure, because it aggregates and becomes denser. The higher the A-CNT percentage, the higher the absorbed Beeswax into the CNT pores, leaving less beeswax on the surface. As seen in Fig. 5 (b), 5 wt.% A-CNT composite still has Beeswax on its surface; meanwhile, on 20 wt.% A-CNT composite shown in Fig. 5 (c), the beeswax is no longer can be seen on the surface, because it was already absorbed into the CNT. This phenomenon indicates the composite ability to maintain its phase. During the melting process, a denser CNT, i.e., 20 wt.% A-CNT composite, the melted beeswax inside the CNT does not leak out as it is

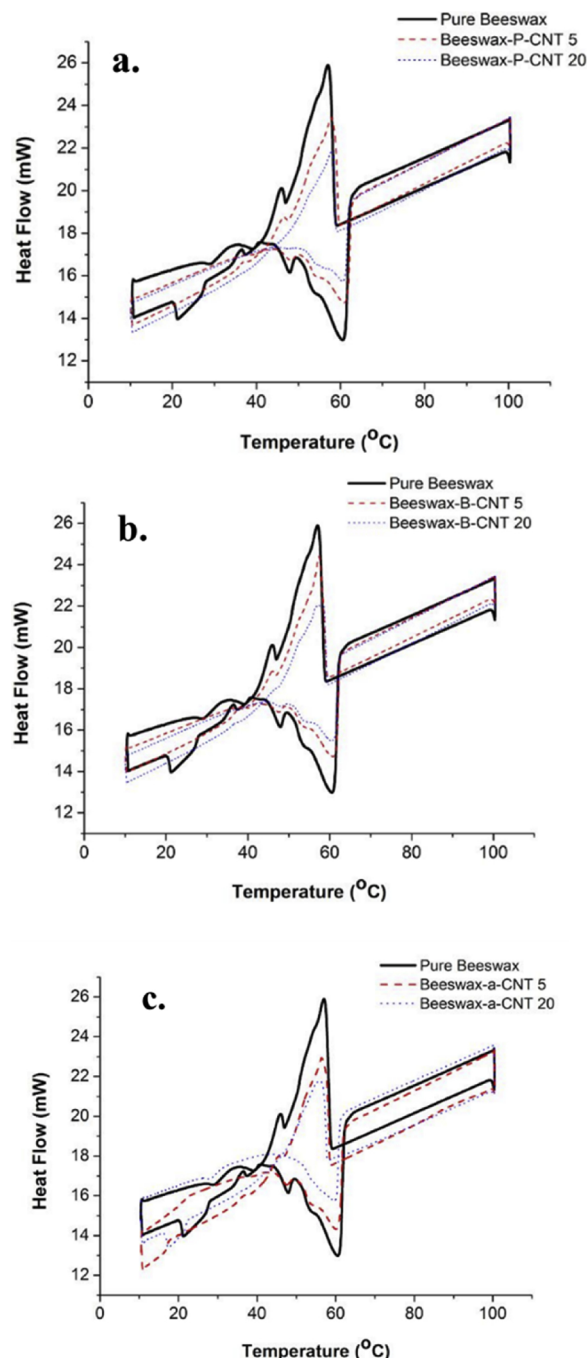


Fig. 7. DSC curves of the obtained composite Beeswax/CNTs with 5 wt% and 20 wt% MWCNT.

- (a) Composite Beeswax/P-CNTs
- (b) Composite Beeswax/B-CNTs
- (c) Composite Beeswax/A-CNTs

Table 2
Phase change temperature and latent heat of the Beeswax and the composite Beeswax/CNTs.

Sample	Heating				Cooling	
	T _m (°C)	Onset (°C)	ΔH (J/g)	T _c (°C)	Onset (°C)	ΔH (J/g)
Paraffin [42]	60.6	52.1	189.4	58.4	60.8	–
Beeswax (Bwx)	60.6	51.4	153.2	57.0	58.7	–154.0
Bwx + P-CNT 20 wt.%	60.4	49.6	90.4	57.6	58.6	–89.9
Bwx + B-CNT 20 wt.%	60.3	49.5	95.9	56.8	59.0	–96.5
Bwx + A-CNT 20 wt.%	59.8	48.9	91.6	55.9	57.7	–91.1
Bwx + P-CNT 5 wt.%	60.6	49.6	131.8	57.4	59.1	–129.7
Bwx + B-CNT 5 wt.%	60.6	48.7	126.0	57.0	58.9	–125.2
Bwx + A-CNT 5 wt.%	60.2	49.4	115.5	56.4	58.5	–126.2

Table 3
Thermal conductivity of composites at T = 40 °C.

Material	K (W/m.K)
Paraffin [42]	0.305
Beeswax [26]	0.25
5 wt.% P-CNT	0.45
5 wt.% B-CNT	0.41
5 wt.% A-CNT	0.46
20 wt.% B-CNT	0.54
20 wt.% A-CNT	0.58

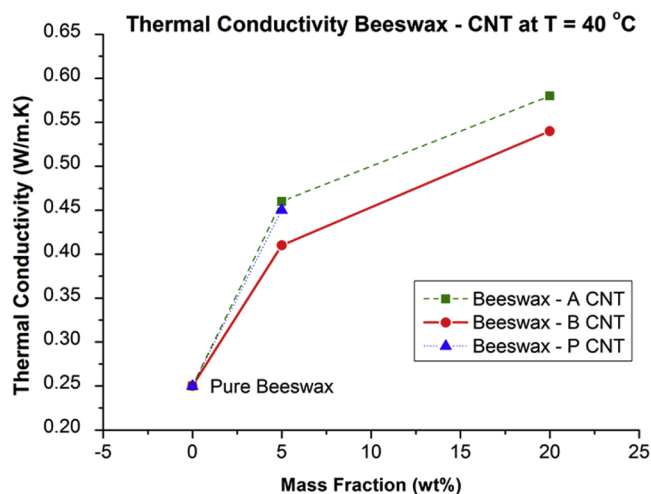


Fig. 8. Thermal conductivity of composites at T = 40 °C.

blocked by the CNT's surface.

3.2. Thermal properties

The thermal properties of pure Beeswax and beeswax/MWCNT composites were analyzed by the differential scanning calorimetry. The melting points, freezing points, characteristic temperatures, and phase-change enthalpies of the samples were examined and compared using the differential scanning calorimetry analysis.

The heating and cooling rates of the DSC were kept at a constant rate of 5°C per minute. All beeswax/MWCNT composite samples were maintained at an initial temperature of 10 °C at the start of the differential scanning calorimetry analysis, and phase-change processes were observed while cycling the temperature back and forth in the ranges of 10–100 °C. The precision of the calorimeter is within 0.2%, and the measurement error for temperature is within 0.05 °C. According to the operational principle of DSC, four characteristic temperatures can be calculated from the heat-flow data: the melting point, onset heating temperature, freezing point, and onset cooling temperature. Fig. 7 shows the DSC analysis of the beeswax/MWCNT nanocomposites

with 5 wt.% and 20 wt.% MWCNTs; the corresponding characteristic temperatures are shown in Table 2.

The initial melting point, heating onset temperature, cooling onset temperature and freezing temperature of beeswax are not changed by the added CNTs. The melting and freezing temperatures were determined as 60.6°C and 57 °C for pure beeswax. Beeswax/MWCNT samples showed trivial deviations in the phase-change temperatures because of the confinement of the Beeswax molecules within the pores of MWCNTs. Shape-stable composites A-CNTs have melting and cooling points of 60.2 °C and 56.4 °C. The melting and freezing temperatures of the beeswax/A-CNT samples are suitable for TES applications in buildings. The slight variation during this phase-change process with different proportions of MWCNTs added is likely due to heat transfer enhancement and the effects of crystallization [41].

The DSC results show the latent-heat values related to melting and freezing were found to be 153.2 J/g and 154.0 J/g for pure beeswax, 115.5 J/g and 126.2 J/g for 5 wt.% A-CNT sample and 91.6 J/g and 91.1 J/g for the 20 wt.% A-CNT sample, respectively. The latent heat clearly decreases with the addition of MWCNTs.

The sensible heat of the composites was determined using Eq. (1) and the DSC experimental data. When the composites phase change from solid to liquid and vice versa, this equation gives the definition of sensible heat.

$$Cp = \frac{\Delta Q}{m \Delta T} \text{ (J/kg } ^\circ\text{C)} \quad (1)$$

Where ΔQ is the heat-flow rate (J/s), m is sample mass (kg), and ΔT is heating rate (°C/s). Figs. 4 and 5 plots the linear equation for sensible heat of heating and cooling, respectively.

Thermal conductivity is significant in the selection of a composite PCM for TES systems, and so enhancing it is an efficient way to make a material suitable for this use [8]. The results in Table 3 list the thermal conductivity of each sample we tested and paraffin, to show the increase gained by adding CNTs. It can be seen that the shape-stable beeswax-CNT samples have higher thermal conductivity than pure beeswax and paraffin.

Fig. 8 plots the thermal conductivities of the samples. Pure beeswax was expected to have a thermal conductivity of approximately 0.25 W/m K at 40°C temperature [26]. The addition of MWCNTs improves the thermal conductivity of the composite. The thermal conductivities of the shape-stable beeswax/A-CNT samples with weight ratios of 20 and 5 wt.% are 0.58 and 0.46 W/m K, respectively. As a result, 20 wt.% of A-CNT increase thermal conductivity by 132% over that of pure beeswax. This shows that MWCNTs can enhance the thermal conductivity of beeswax.

3.3. Thermal stability

The main concern in TES materials is the stability of the thermal and chemical properties for long-term installation. Consequently, a thermal-cycling test was implemented to examine properties of composite PCM after many melting and freezing cycles. The thermal-cycling tests used

Table 4
Phase change temperature and latent heat after thermal cycle of the Beeswax and the composite Beeswax/CNTs.

Sample	Cycles	Heating			Cooling		
		T _m (°C)	Onset (°C)	ΔH (J/g)	T _c (°C)	Onset (°C)	ΔH (J/g)
Beeswax	0	60.6	51.4	153.2	57.0	58.7	−154.0
	150	60.4	51.6	119.9	57.1	58.8	−132.8
	300	60.0	61.3	122.4	56.7	58.3	−138.3
Beeswax / A-CNT 5 wt. %	0	60.2	49.4	115.5	56.4	58.5	−126.2
	150	59.4	51.3	106.3	56.0	57.6	−116.7
	300	58.2	49.4	97.9	55.7	57.5	−109.0

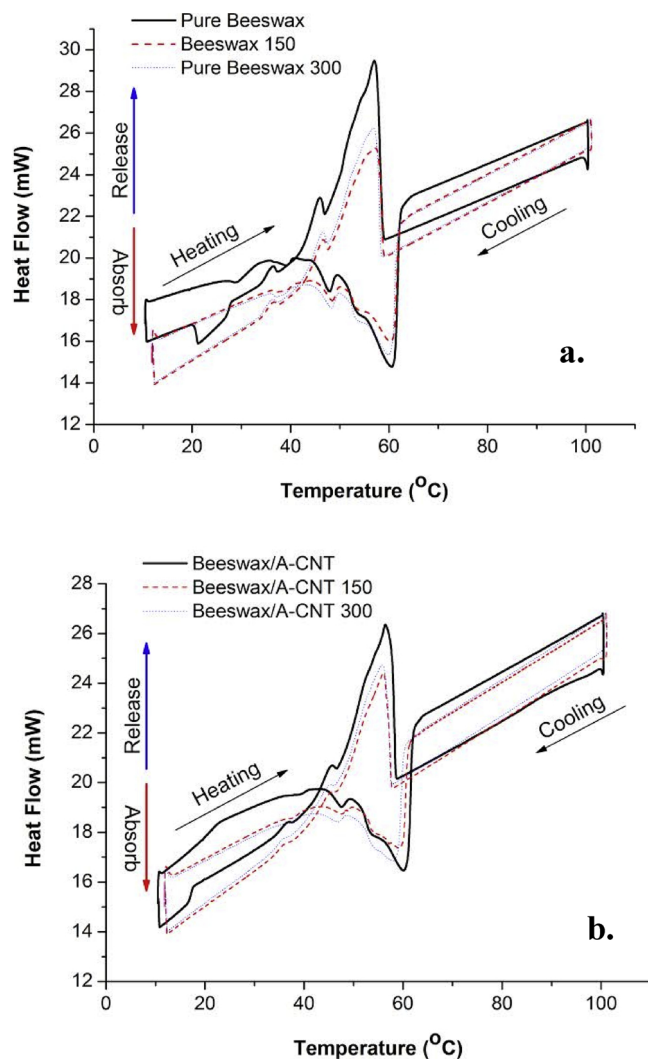


Fig. 9. DSC curves after thermal-cycle testing: (a) Pure Beeswax (b) Beeswax/A-CNT 5 wt. %.

300 cycles and DSC analysis was performed after 0, 150, and 300 cycles.

Thermal properties of SSPCM before and after thermal cycling are given in Table 4. After thermal cycling, the melting temperature of 5 wt. % beeswax/A-CNT after 150 and 300 cycles changed by 0.8 and 2.0 °C, and the freezing temperature changed by 0.4 and 0.7 °C, respectively. The latent-heat values of the composite PCM were changed by 8% and 15.2% for melting process and 7.5% and 13.6% in the freezing process after 150, and 300 cycles as reflected in Fig. 9. The latent heat of composite declines less than that of pure beeswax. Moreover, the changes in melting and freezing temperature of the composite PCM are

not significant.

By analyzing the temperature evolutions during charging and discharging, the addition of MWCNTs has a strong effect on the heat transfer properties of beeswax in both solid and liquid states. As a result of the combined effects of the improvement of thermal conductivity and the restraint of natural convection by the MWCNTs, the charging and discharging rate of 5 wt. % beeswax/A-CNTs was faster than pure beeswax. Furthermore, the freezing process is faster than the melting process for all of the samples. This phenomenon indicates a difference in the heat transfer properties within the nanocomposite during the melting and freezing processes [41].

4. Conclusions

The preparation of beeswax-CNT composites and their physical, chemical, and morphological properties for thermal storage were discussed in this paper. Three types of MWCNTs were modified and tested in order to prepare shape-stable composites with the addition of P-CNTs, B-CNTs, and A-CNTs. The major conclusion of this study are:

- (1) Experiments showed that the SSPCM was successfully fabricated only with impregnation of A-CNTs into Beeswax. Its thermal conductivity was increased when using CNTs in different mass fractions of 5% and 20%. Visual inspection and FTIR and SEM analysis confirm the good physicochemical compatibility of A-CNTs for use as the supporting material for composites.
- (2) DSC results indicate that the prepared Beeswax/A-CNTs 5 wt. % and Beeswax/A-CNTs 20 wt. % composite PCMs had melting temperatures of 60.2 °C and 59.8 °C, and latent heat values of 115.5 J/g and 91.6 J/g, respectively.
- (3) The Beeswax/A-CNT 5 wt. % composite PCM sample exhibited good long-term TES reliability and shape stability. Additionally, with the addition of CNTs in a mass fraction of 20 wt. %, the thermal conductivity of the composite PCM increased significantly, by 132%. The total melting and freezing times of the composite PCMs with 5 wt. % CNTs were significantly less than the other sample because of the enhanced thermal conductivity.
- (4) The composite PCM with Beeswax/A-CNTs 5 wt. % shows promise as a TES material for thermal-regulation applications in buildings and greenhouses. The specific material for an application will depend on the climatic conditions of the location as well as on the recovery of excess heat discharged during industrial processes.
- (5) As the first work done in preparation and characterization of beeswax/multi-walled carbon nanotubes PCM for thermal energy storage, the result is very encouraging. However, the large-scale TES performance and optimal conditions for this composite PCM remain to be tested in further research.

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